Section 4.1 — Exponential Functions

Math 103: College Algebra

Identifying Exponential Expressions

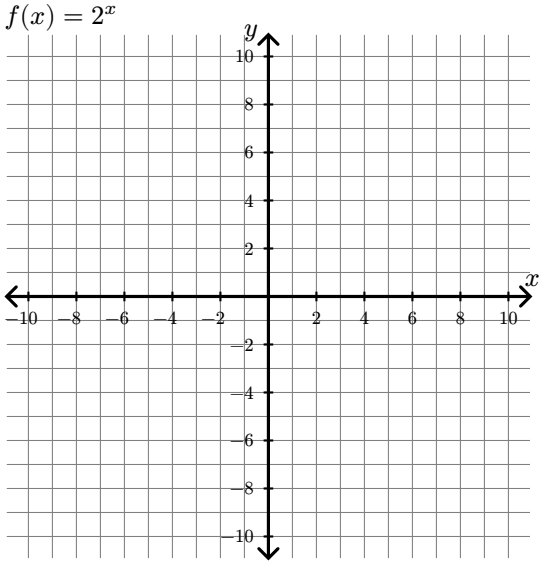
Determine whether or not each expression is exponential.

1. 8^x
2. $x^{3/2}$
3. $\left(\frac{2}{3}\right)^{3x-4}$
4. $3 \cdot 4^x - 3^{x^2}$
5. $x^2 - 4x + 3$

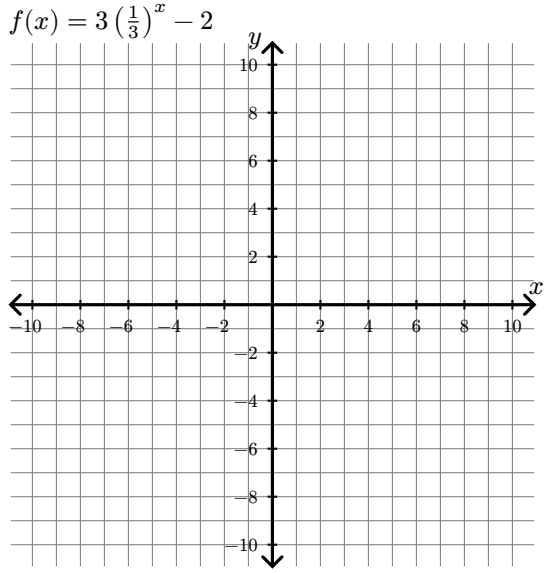
Graphing Exponential Functions

For each exponential function, find the points on the graph for each input (not x-value) in $\{-2, -1, 0, 1, 2\}$, and find the equation of the horizontal aysmptote. Then, sketch a graph of the function in the space provided. (If the point can't be plotted in this space, leave it off. Do your best to approximate fractions.)

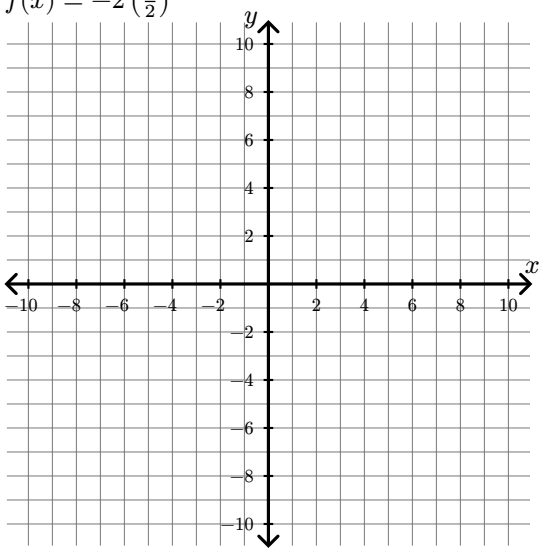
6. $f(x) = 2^x$



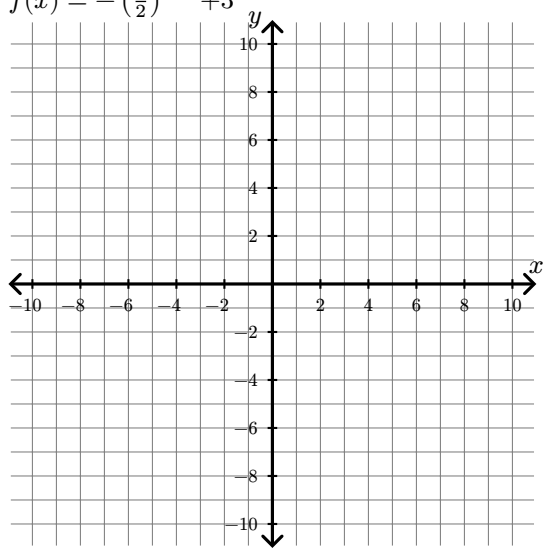
7. $f(x) = 3\left(\frac{1}{3}\right)^x - 2$



8. $f(x) = -2\left(\frac{3}{2}\right)^{x-1}$



9. $f(x) = -\left(\frac{1}{2}\right)^{x+1} + 3$

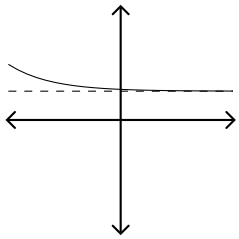


Identifying Graphs of Exponential Functions

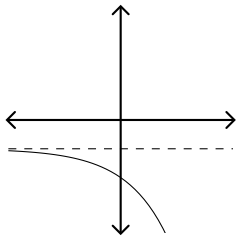
Match each function to its graph, and give the equation of its horizontal asymptote.

10. $f(x) = -\left(\frac{2}{3}\right)^{x+1} - \frac{1}{3}$
11. $g(x) = \left(\frac{1}{2}\right)^{x+4} + 1$
12. $h(x) = \left(\frac{11}{4}\right)^{x-1} - \frac{3}{2}$
13. $k(x) = -2^x - 1$

a.



b.



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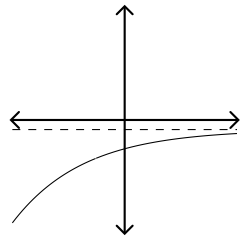
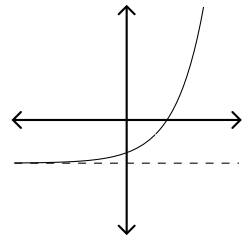


d.



Domain and Range of an Exponential Function

Find the domain and range of each exponential function.

14. $f(x) = 2^{x+3} - 1$
15. $g(x) = -\left(\frac{1}{2}\right)^{x-5} - 11$
16. $h(x) = -10^{x+1} + 1$
17. $k(x) = (0.23)^x + 623$

Exponential Equations

Solve each equation.

18. $3^x = 81$
19. $27^x = 243$
20. $25^{x+1} = \frac{1}{5}$
21. $3^{x^2-2x-4} = \frac{1}{3}$
22. $4^{2x-1} = 32^{x+2}$
23. $\left(\frac{1}{2}\right)^x = 16$
24. $8^{2x^2} = 4^{12x-9}$
25. $\left(\frac{1}{3}\right)^{x^2} = 9^{4-3x}$